

# GAHAN SARAIYA

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## OBJECTIVE

Senior Software Engineer with 8+ years of experience, currently at AMD India Pvt Ltd managing AI/ML workload execution on HPC GPUs. Strong command of Python, HPC, UEFI firmware, DevOps, and cluster observability (Prometheus, Grafana, Slurm). Experience spanning Agentic AI, knowledge-base design, and CI/CD for large-scale GPU clusters. Open-source contributor focused on practical system software and parallel computing.

## SOFTWARE TOOLS

|                     |                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Languages           | Python, C, C++, Shell Scripting, SQL, LaTeX, YAML                                                                                             |
| Frameworks          | Tensorflow, Pytorch, LangGraph, FAISS, TensorRT, Pytest, Edk2, PyCUDA, Pandas, UEFI, Selenium, Redfish                                        |
| Architecture Design | Parallel Programming, ROCm, HIP, Cluster Observability, Agentic AI                                                                            |
| Tools and Services  | GitHub Actions, Claude, Cursor, LLM APIs, Restful APIs, Teamcity, Terraform, Docker, Docker Compose, Jenkins, Git, Prometheus, Grafana, Slurm |
| Cloud Technologies  | GCP, AWS (S3, EC2, SageMaker), Azure, Vertex AI, Digital Ocean, JFrog Artifactory, TeamCity, JIRA                                             |
| Observability & Ops | Telemetry, Log Shipping to S3, Observability Agents, Cluster Monitoring                                                                       |
| System Level IP     | RAS, Compute Partitioning, Boot time optimization, CPER                                                                                       |

## EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |
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| Senior System Software Designer at AMD India Pvt Ltd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mar 2024 - Present   |
| <ul style="list-style-type: none"><li>• Manage cluster observability and telemetry across large GPU networks using Prometheus, Grafana, and Slurm; automate deployment of observability runners and agents.</li><li>• Design Obsidian-based agentic memory network for agent memory index, knowledge base, and skills auto-sync (human-readable and agent-readable).</li><li>• Automate cluster validation CI with secure telemetry log shipping to AWS S3.</li><li>• Conduct functional and performance benchmarking of HPC GPUs with test suites aligned to industry standards (TensorFlow, PyTorch, GEMM).</li><li>• Define load-balanced GPU performance criteria covering thermal limits, power, memory usage, and core utilization.</li><li>• Document RAS injection procedures for MI2XX and MI3XX platforms; cover partitioning, SMI, HIP profiling, and parallel execution.</li><li>• Collaborate with engineers and researchers on HPC and Agentic AI tooling.</li></ul> |                      |
| Senior Firmware Development Engineer at Intel Technology India Pvt Ltd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Oct 2023 - Mar 2024  |
| <ul style="list-style-type: none"><li>• Platform provisioning and infrastructure setup for mass automation of HPC products.</li><li>• Platform Orchestration Layer design of Universal Scalable Firmware (USF).</li><li>• Harnessing Azure OpenAI Generative AI adaptation for Firmware Development Productivity for assessment code review, guidance, basic security checks of code.</li><li>• GitHub Action and JIRA integration for managing Continuous deployment and test through CloudBees Jenkins with baremetal and docker nodes.</li><li>• Enhancing firmware-based boot-time performance for notebook devices.</li></ul>                                                                                                                                                                                                                                                                                                                                                 |                      |
| Firmware Development Engineer at Intel Technology India Pvt Ltd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Jun 2020 - Sept 2023 |

- Lead the next generation client platform and architecture design at USF POL alongside the domain ownership for Automation Framework, Platform Provisioning, Applied Generative AI & Prompt Engineering and Quantum Firmware Design.
- The end-to-end Software Development Life Cycle is managed, which includes licensing, legals, and IP planning, as well as applicable Code Security and Static Code Analysis.
- Jenkins configuration for Security tools - Snyk, Coverity
- Improving github code review hooks for python packaging and linting through bandit and flask as Jeninks pipeline
- automated testing of firmware functional validation of Firmware with TeamCity.
- Mentoring and knowledge development engagement activities.

#### **Firmware Engineer Intern at Intel Technology India Pvt Ltd**

Jun 2019 - May 2020

- BIOS Firmware Development for Client products with the framework EDK2, UEFI Firmware, Involvement with assessing new debugging solutions and OpenBoard Package

#### **Teaching Assistantship at Nirma University**

Aug 2018 - May 2019

- Data Mining and Visulaization

#### **Software Developer at Quixom Technology**

Jul 2017 - Apr 2018

- SaaS-Employee management system (chat, surveys, benchmark, analytics...) and Restful API Development
- Socket programming in python to receive continuous data from IoT device and display chunks of data meaningfully
- Streaming service with Web-Portal, DRF and Kodi across platforms

#### **Freelance Full-stack Developer**

2015 - Apr 2019

- Cloud solution for Data gathering and webscraping with selenium and GCP.
- Designing template and banners for High resolution prints
- Inventory management system for Book store with Django.

## **PROJECTS**

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#### **Cluster Management and Observability**

2025 - Present

**Description:** Cluster management and observability for large-scale GPU networks at AMD.

- Integrate industry-standard telemetry stacks (Prometheus, Grafana, Slurm) for cluster network monitoring.
- Auto-deploy observability runners and agents across large cluster fleets.
- Automate secure log upload and telemetry shipping to AWS S3 with CI for cluster validation.

#### **Obsidian Agentic Memory Network**

2025 - Present

**Description:** Agentic design for memory index, knowledge base, and skills sync readable by both humans and agents.

- Deliver Obsidian-based agentic memory network with indexed knowledge base and auto-synced skills for optimal human and agent consumption.

#### **Agentic AI for Software Engineering Modernization**

2025 - Present

**Description:** Modernized enterprise workflows, repository management, and code review using multi-tool Agentic AI (Claude, Cursor).

- Built tools, automated git hooks, and modular agents with security compliance and token spend controls.

## Context-Aware Task Routing & Triage with LangGraph

2025 - Present

**Description:** Context-aware task/issue routing and resolution using graph-based LLMs.

- Designed routing with LangGraph and FAISS semantic indexes; integrated OpenAI, Vertex AI, and local LLM APIs.
- Built connectors for GitHub, Jira, and historical systems with a persona counselor for summarization.

## GPU Performance Analysis for Data Center Workloads

2024 - Present

**Description:** Solution design for in-band profiling through System Management Interface

- **Architected In-Band Profiling Solutions:** Designed and implemented robust in-band profiling mechanisms utilizing the System Management Interface (SMI) for comprehensive GPU telemetry.
- **Dynamic Workload Orchestration:** Developed systems for dynamic workload creation, intelligently adapting to real-time hardware resource availability (e.g., GPU utilization, memory) to optimize testing and analysis.
- **Comprehensive Hardware Metric Evaluation:** Engineered processes for capturing and analyzing critical hardware metrics, including GPU temperature, power consumption, and resource utilization, directly through SMI and in-band channels.
- **Granular Program Performance Analysis:** Implemented detailed program profiling techniques, including the capture and analysis of kernel call stack timings, to pinpoint performance bottlenecks and inefficiencies.
- **Advanced Analytical Metric Design:** Innovated a methodology to map multiple disparate key factors (e.g., thermal, power, utilization, kernel timings) into a unified, insightful data metric, significantly enhancing the depth and clarity of system performance analysis.

## Prompt Engineering and Generative AI for firmware stack

2023 - 2024

**Description:** Solution design for accelerated silicon tape-in with accelerating firmware deliveries

- Work task summarization with Azure LLM Open AI services (GPT3.5 and GPT4)
- Code Review against standard static code analysis and common vulnerabilities finding
- Generation of Code Enhancement suggestion

## Platform Provisioning with Pre-Production Reference Board

2023 - 2024

**Description:** Platform Provisioning with automating most common features for pre-production server processor such as:

- Remote BMC and BIOS Firmware Flashing
- OS Deployment from artifactory through Jenkins Pipeline
- Boot to EFI Shell and OS
- Configuration of Initial system setup with Jenkins pipeline steps with Baseboard management controller RedFish API

## Universal Scalable Firmware

2022 - 2024

**Description:** Architecture design of Platform orchestration layer for Firmware Configuration

- Motivation of workgroup - [Firmware Configuration – Past, Present, and Future](#)
- [Firmware Configuration](#)

## Firmware Automation Framework

2020 - Present

**Description:** Lead SDL for an automation foundation framework for high scaled volume of silicon devices with edk2 UEFI firmware;

- Seamless configuration of system without any manual intervention for BIOS configuration.

- SDL activities like legal compliance, security, licensing, static code analysis
- Continuous deployment and test with jenkins and tox configurations
- Release Artifact management to JFrog Artifactory using GitHub Actions
- <https://github.com/intel/xml-cli>

## Multiview Video Summarization

2019 - 2020

**Description:** Summarizing video of multiple view angles of egocentric location in to one to generate effective video summary, minimizing overhead to explore multiple cam feed to discover accidents or anomalies.

## EDUCATION

### M.Tech Computer Science

Jul 2018 - Jun 2020

Nirma University, GPA: 8.49

### B.E. in Computer Science and Engineering

Aug 2013 - Apr 2017 Gujarat Technological University,

CGPA: 8.08

### 12th Science, GSHSEB, Class XII

Jun 2011 - Mar 2013

Science Stream, 74.92%

## CERTIFICATIONS

|      |                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------|
| 2025 | <a href="#">AMD ROCm Star - Application Developer</a> by AMD Adaptive Computing Training Team                                   |
| 2025 | <a href="#">On-Device AI</a> by Qualcomm & DeepLearning.ai                                                                      |
| 2025 | <a href="#">Essential Terraform in AWS</a>                                                                                      |
| 2021 | <a href="#">Product Assurance and Security Yellow Belt - Software</a>                                                           |
| 2020 | Natural Language Processing with Classification and Vector Spaces<br><a href="#">Coursera S9DGGMCDMBGQ</a>                      |
| 2020 | Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization<br><a href="#">Coursera SMWQJKJYJW6G</a> |
| 2020 | Structuring Machine Learning Projects - <a href="#">Coursera SEJML7H3TN5Z</a>                                                   |
| 2020 | Neural Networks and Deep Learning - <a href="#">Coursera S2TETZHTLUSL</a>                                                       |
| 2020 | Machine Learning - <a href="#">Coursera XVNXNPPZ3JMA</a>                                                                        |

## ACHIEVEMENTS

|              |                                                                                            |
|--------------|--------------------------------------------------------------------------------------------|
| 2026         | Top 1% recognition for AI use-case development                                             |
| 2026         | Top 10 in AI Hackathon: Hack2Skill Prompt Wars by Google (In-person, Bangalore)            |
| 2026         | Rank #3 in Virtual Prompt Wars Hackathon                                                   |
| 2023         | Firmware Configuration Presentation at UEFI Developers Conference & Plugfest               |
| 2023         | Tech Talk on Harnessing Multiprogramming across CPU, GPU and QPU (Quantum Processing Unit) |
| 2021         | Expert Lecture on CUDA Programming                                                         |
| 2019-Present | 4000+ reputation on Stackoverflow                                                          |
| 2017-Present | Active volunteer on Peer Learning, Cultural and team building activity                     |
| 2018         | Rank #1 for Python challenges on hackerrank                                                |
| 2018         | Rank #112 from 2 lacs+ participants in Techgig's National Coding Contest 2018              |

## PUBLICATIONS

|                                                                                                      |      |
|------------------------------------------------------------------------------------------------------|------|
| • <a href="#">HIP Parallel Programming with ROCm</a> Lecture Series, YouTube                         | 2025 |
| • <a href="#">Beyond the Power — Data Center &amp; BMC</a> Lecture Series, YouTube                   | 2025 |
| • <a href="#">System Design for Parallel Programming</a> Lecture Series, YouTube                     | 2025 |
| • <a href="#">Firmware Configuration – Past, Present, and Future.</a> Presentation, UEFI DevCon 2023 | 2023 |